

Facial Expression Recognition: Emotion Classification

Course Computer Vision, HSLU

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Date 09 June 2026

Dataset FER-2013 (Kaggle: <https://www.kaggle.com/datasets/msambare/fer2013>)

Model ResNet18 — Transfer Learning

Baseline CLIP Zero-Shot (ViT-B/32)

GitHub Project Documents and Full Notebook with Code:
https://github.com/noiiram/CV_FS2026_Emotions

1. Introduction

Facial expression recognition (FER) is a core task in affective computing with applications in human-computer interaction and behavioural analysis. Automatically classifying emotions such as happiness, sadness, anger, or surprise from facial images is challenging due to variability in lighting, pose, and individual appearance, and the inherent ambiguity of facial expressions.

This project investigates two complementary approaches on the FER-2013 dataset:

1. **Supervised fine-tuning** of a ResNet18 backbone pre-trained on ImageNet, using transfer learning to adapt generic visual features to emotion recognition.
2. **Zero-shot classification** using CLIP (Contrastive Language-Image Pretraining), which requires no task-specific training and instead leverages natural language prompts to classify images.

The trained models are evaluated quantitatively (accuracy, confusion matrix, per-class F1) and qualitatively through feature map visualisation and out-of-distribution testing on real photos of the project team members.

2. Setup

The environment is configured with the necessary python libraries for deep learning (PyTorch, torchvision), data handling (NumPy, PIL), and visualisation (Matplotlib). Training is performed on GPU where available, falling back to CPU otherwise. This is verified at startup via the function `torch.cuda.is_available()`.

2.1 Load Kaggle Dataset

The FER-2013 dataset is downloaded directly from Kaggle using the Kaggle API and extracted into a local `fer2013/` directory. The dataset is pre-structured into `train/` and `test/` subfolders, each containing one subfolder per emotion class, compatible with PyTorch's `ImageFolder` format.

3. Data exploration

The FER-2013 dataset (<https://www.kaggle.com/datasets/msambare/fer2013>) contains 35,887 grayscale images of size 48×48 pixels. Each image is labelled with one of seven basic emotions: *angry*, *disgust*, *fear*, *happy*, *neutral*, *sad*, *surprise*. Images were collected by querying Google Image Search and are therefore highly heterogeneous in terms of lighting, pose, occlusion, and image quality. We first inspect the structure of the FER-2013 dataset and visualise one sample image per emotion class.

Classes found: ['angry', 'disgust', 'fear', 'happy', 'neutral', 'sad', 'surprise']

Dataset structure:

TRAIN:

angry	:	3995 images
disgust	:	436 images
fear	:	4097 images
happy	:	7215 images
neutral	:	4965 images
sad	:	4830 images
surprise	:	3171 images
TOTAL	:	28709 images

TEST:

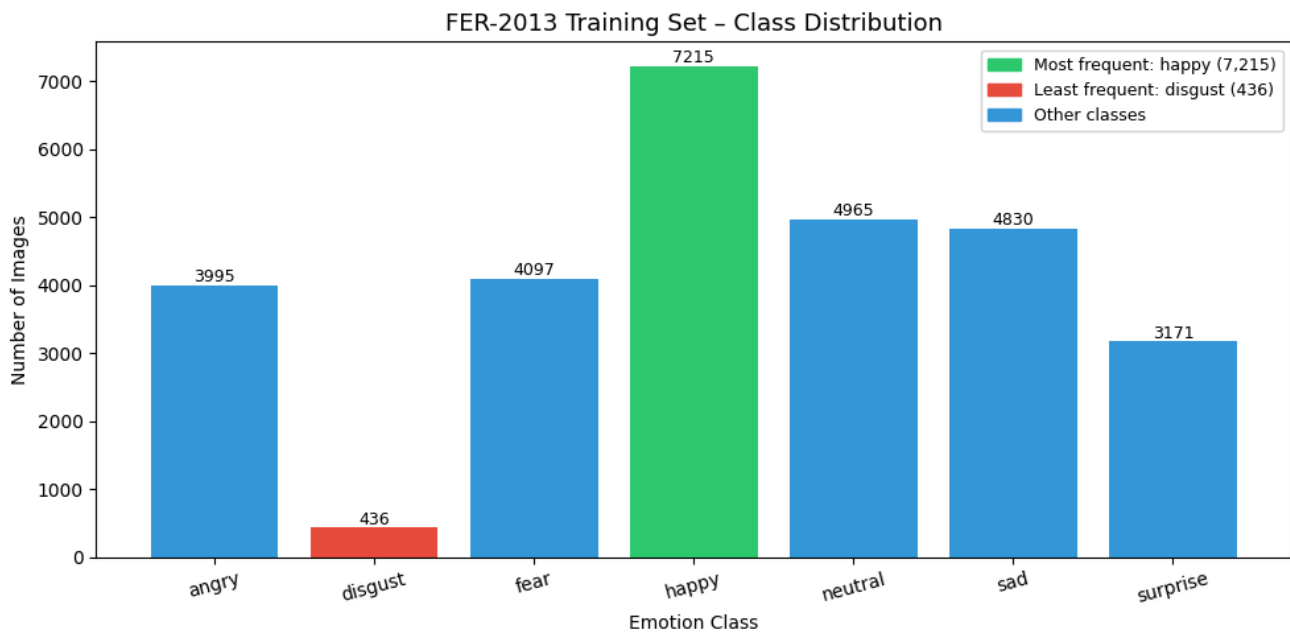
angry	:	958 images
disgust	:	111 images
fear	:	1024 images
happy	:	1774 images
neutral	:	1233 images
sad	:	1247 images
surprise	:	831 images
TOTAL	:	7178 images

FER-2013: Sample Images per Class



3.1 Class Imbalance Analysis

Analysis of the data from the sample images reveals that the number of images per emotion varies. This can lead to the model tending to predict the more frequently occurring classes. We address this issue through data augmentation during the pre-processing stage (Chapter 4).



Imbalance ratio (max/min): 16.5x

3.2 Data Cleaning

When analysing the data from the sample images, it was also noted that Watermarks are visible in the sample images. We want to remove these before training so that they do not interfere with the training process. The dataset is scanned for images that contain visible watermarks or text overlays. To detect watermarked images automatically, we combined two complementary methods into an ensemble detector.

- **CLIP-based detection** compares each image against two prompts: "*a photo with a visible watermark*" vs. "*a clean photo without any watermark*" and returns a probability score. CLIP captures high-level semantic cues about the presence of overlaid text, but can be fooled by bright or high-contrast image regions that superficially resemble watermarks.
- **Pixel-based detection** complements CLIP with a low-level heuristic: it identifies pixels that are simultaneously very bright (intensity > 200) and exhibit high local variance (computed over a 5×5 neighbourhood). Semi-transparent text overlays characteristically produce this pattern.

An image is only flagged if both signals agree, minimising false positives. If CLIP returns nan, only the pixel-based approach is used. Flagged images are moved to a backup folder (not deleted) so the operation is reversible.

The CLIP scores all range between 0.53 and 0.99. This means that CLIP identifies almost every image as a watermark. Whilst this is not surprising given this dataset, it is not particularly useful. The pixel score, on the other hand, works well. The final pipeline therefore relies exclusively on the pixel-based detector.

Top 20 suspected watermarked images — verify before removing



Flagged images by emotion class:

```
angry      : 2631
disgust    : 302
fear       : 3287
happy      : 4584
neutral    : 3172
sad        : 2899
surprise   : 2988
```

4. Preprocessing & DataLoaders

FER-2013 images are 48×48 grayscale. The following preprocessing pipeline is applied:

- Grayscale → 3-channel RGB:** Required by the ResNet18 architecture, which expects three input channels.
- Resize to 224×224:** Matches the input resolution expected by ImageNet-pretrained backbones.
- Data augmentation (training only):**
 - Random horizontal flip
 - Random rotation $\pm 10^\circ$
 - Colour jitter (brightness and contrast $\pm 30\%$)
- ImageNet normalisation:** Mean `[0.485, 0.456, 0.406]`, std `[0.229, 0.224, 0.225]`.
Applied because the backbone was pre-trained on ImageNet; using the same normalisation ensures the pre-trained feature representations remain valid.

Augmentation is applied only to the training set to avoid data leakage into validation and test evaluation.

5. Model Architecture — ResNet18 + Transfer Learning

We use **ResNet18** as our backbone. ResNet introduced residual connections (skip connections) that allow gradients to flow directly through the network without vanishing, enabling stable training of deeper architectures. ResNet18 consists of four residual blocks (`layer1` – `layer4`), preceded by an initial convolution and followed by global average pooling and a fully connected classification head.

5.1 Transfer Learning Strategy

Transfer learning is the practice of reusing a model pre-trained on a large dataset (here: ImageNet, 1.2M images) for a different but related task. The rationale is that low-level features — edges, textures, colour gradients — are universal and task-independent, while high-level features are task-specific.

Our strategy:

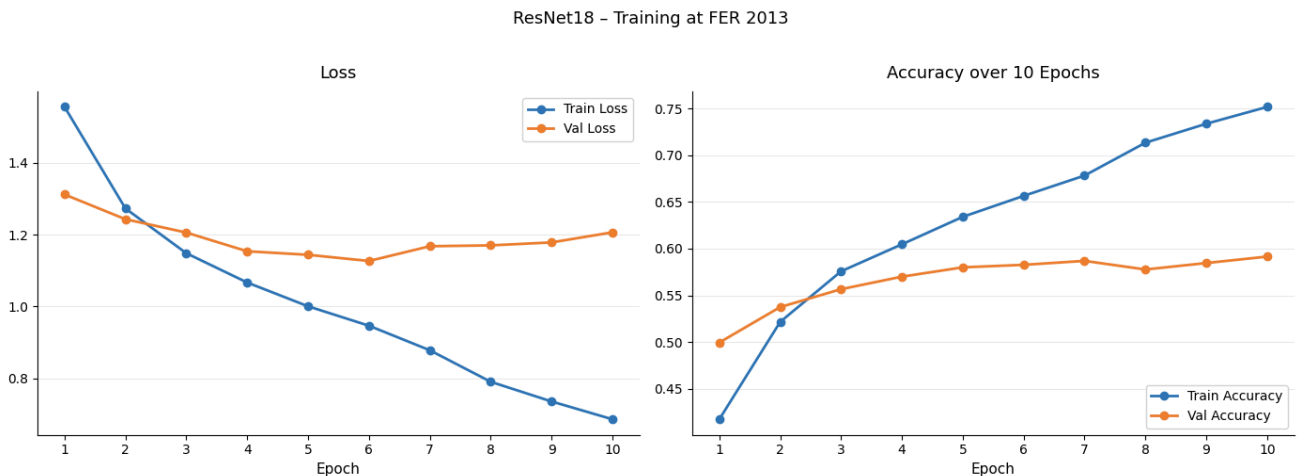
- **Freeze** `conv1` , `bn1` , `layer1` , `layer2` , `layer3` : These layers encode generic ImageNet features that transfer well to facial images without modification.
- **Fine-tune** `layer4` and the classification head: The deepest convolutional block is adapted to FER-2013-specific features, while a new fully connected layer (`512` → `7`) replaces the original ImageNet head.
- **Dropout(0.5)** is inserted before the final linear layer to regularise the model and reduce overfitting on the relatively small FER-2013 training set.

This selective fine-tuning strategy reduces training time significantly (fewer parameters updated), prevents overfitting, and leverages the representational power of ImageNet pre-training.

6. Training

Training is conducted in two phases: an initial 10-epoch run, followed by a further 10-epoch extension to allow additional convergence. The model was trained using cross-entropy loss, which is the standard choice for multi-class classification problems as it penalises confident wrong predictions more heavily than uncertain ones. As optimiser we used Adam with a weight decay of $1e-4$, which applies L2 regularisation to penalise large weights and reduce overfitting. The learning rate was set to $1e-4$ and further controlled by a ReduceLROnPlateau scheduler with a reduction factor of 0.5 and a patience of 2 epochs — meaning the learning rate is halved whenever the validation loss shows no improvement for two consecutive epochs, allowing finer convergence in later stages of training. Training was conducted with a batch size of 64. Throughout the entire training process, the model checkpoint achieving the highest validation accuracy was saved and used for all downstream evaluation, ensuring that the reported results reflect the best generalisation performance rather than the final epoch state

6.1 Training Curves



The loss and accuracy curves reveal a classic overfitting pattern. The training loss decreases continuously across all ten epochs, dropping from approximately 1.55 to 0.68, while the validation loss initially follows this trend until epoch 4, then stagnates around 1.15 and slightly increases towards epoch 10. This divergence is a clear indicator that the model begins memorising the training data rather than learning generalisable features.

The accuracy curves confirm this observation. Training accuracy rises steadily throughout all ten epochs, reaching approximately 75% by epoch 10. Validation accuracy improves rapidly in the first two epochs — from 50% to around 54% — then slows considerably and plateaus at approximately 59% from epoch 5 onwards, resulting in a train-validation gap of roughly 16 percentage points by the end of the first training phase.

The best checkpoint is therefore saved at around epoch 4–5, where validation accuracy reaches its peak before stagnating. The applied regularisation measures — Dropout(0.5) and weight decay — reduce but do not fully eliminate the overfitting, which is expected given the limited size and inherent label noise of FER-2013. Based on this analysis, a second training phase of 10 additional epochs is conducted to verify whether the validation plateau is stable or whether further convergence is possible.

6.2 Extended Training

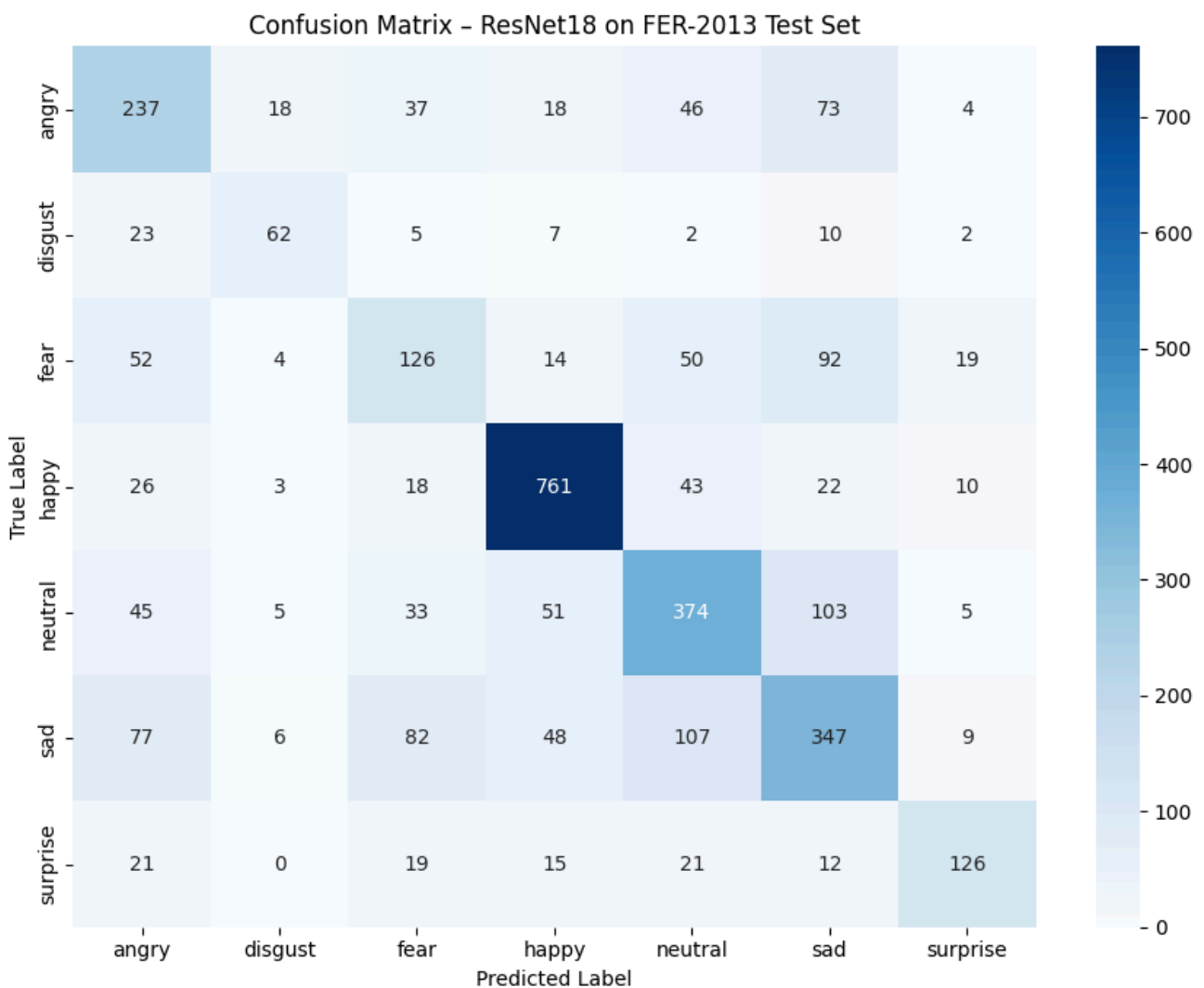
Training is continued for a further 10 epochs to verify that the validation plateau is stable and not a temporary local minimum.

7. Evaluation and Confusion Matrix

We evaluate the best saved model on the held-out test set. In addition to overall accuracy, we report a full classification report (precision, recall, F1 per class) and a confusion matrix to identify systematic failure modes.

Test Accuracy: 0.6179 (61.8%)

	precision	recall	f1-score	support
angry	0.49	0.55	0.52	433
disgust	0.63	0.56	0.59	111
fear	0.39	0.35	0.37	357
happy	0.83	0.86	0.85	883
neutral	0.58	0.61	0.59	616
sad	0.53	0.51	0.52	676
surprise	0.72	0.59	0.65	214
accuracy			0.62	3290
macro avg	0.60	0.58	0.58	3290
weighted avg	0.62	0.62	0.62	3290



7.1 Feature Map Visualisierung

We compare feature maps for three different emotion classes to analyse whether the network develops class-specific representations in deeper layers.

CNNs learn hierarchical features:

- Early layers (e.g. layer1) detect low-level patterns such as edges, corners, and colour gradients — similar across all images.

- Deep layers (e.g. layer4) detect high-level, abstract structures specific to the task — ideally capturing emotion-relevant facial features.

Early Layer (layer1)

The 16 feature maps in layer1 all still clearly show the face as a structure. Each map highlights different aspects — one emphasises the horizontal lines around the mouth and eyes, another reacts strongly to high-contrast edges along the facial contour, and yet another shows the texture of the skin. This is exactly the expected behaviour: layer1 learns general, local image filters similar to classical edge detectors (Sobel, Gabor). The activations are dense — almost every pixel has a value, because edges and textures occur throughout the image.

Deep Layer (layer4)

Here, the picture is completely different. The feature maps are much coarser (lower spatial resolution due to pooling) and significantly more selective — the yellow (high activation) is concentrated in a few specific regions. In this 'happy' image, you can see activations concentrated around the mouth area, which makes sense: a smile is the strongest visual signal of happiness. Other maps show activations around the eye area. The dark (purple) areas indicate no activation or negative activation — the filter does not respond there at all.

8. CLIP Zero-Shot Baseline

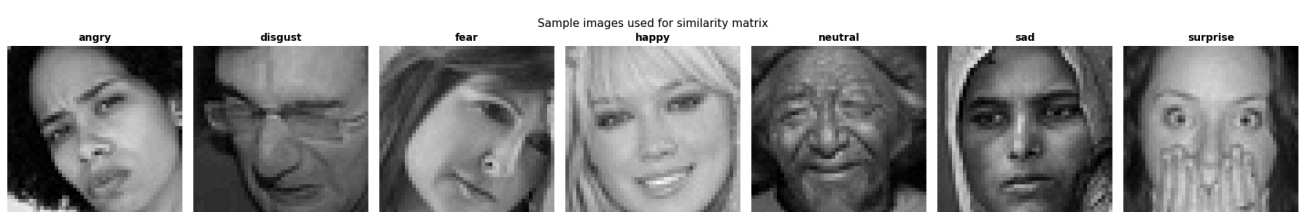
CLIP (Contrastive Language-Image Pretraining) is a vision-language model trained on 400 million image-text pairs via contrastive learning. It learns a shared embedding space where semantically related images and texts are positioned close together. At inference time, zero-shot classification proceeds as follows:

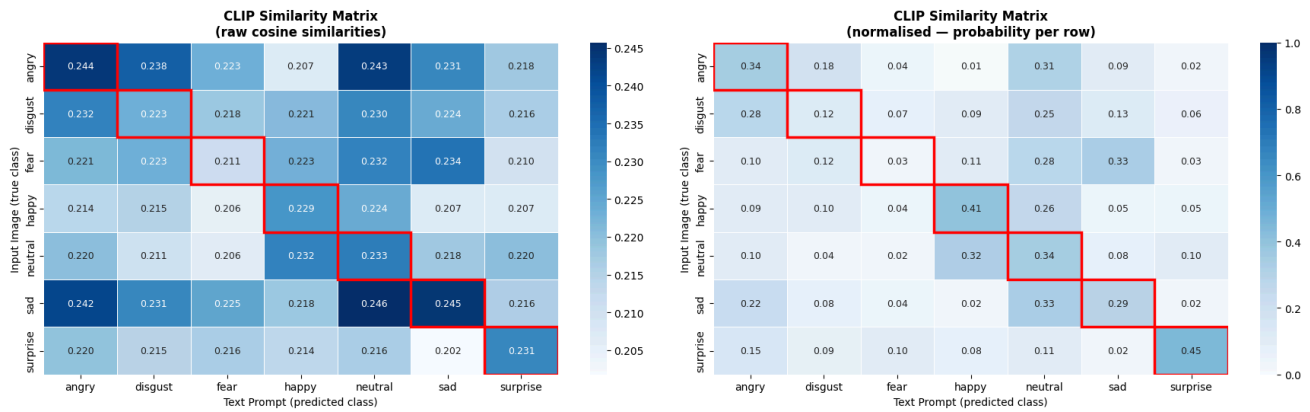
1. The input image is encoded into an embedding vector by the CLIP image encoder.
2. For each candidate class, a text prompt is constructed — e.g. "a photo of a person with happy expression" — and encoded by the CLIP text encoder.
3. Cosine similarity between the image embedding and each text embedding is computed. The class with the highest similarity is the prediction.

This requires **no labelled training data** and serves as a strong upper bound for what general-purpose vision-language knowledge can achieve on FER-2013 without task-specific adaptation.

8.1 CLIP Similarity Matrix

Visualising this matrix makes the CLIP zero-shot mechanism directly visible.





The seven sample images shown above represent the specific images used for this analysis. All selected images show clearly recognisable facial expressions, making this a representative qualitative evaluation.

The raw cosine similarities (left matrix) cluster tightly between 0.205 and 0.245 across all classes, confirming that CLIP operates with low confidence on 48×48 grayscale images.

After row-wise normalisation (right matrix), the class structure becomes clearer. Surprise (0.45) and happy (0.41) show the strongest diagonal values, benefiting from visually distinctive features such as raised eyebrows, open mouth, and a clear smile. Neutral is correctly predicted at 0.34, though the adjacent happy score of 0.32 reflects the ambiguity between these two classes. Fear performs worst with a diagonal value of only 0.03 — CLIP assigns the image to sad (0.33) and neutral (0.28) instead. Disgust is misclassified as angry, and sad is predicted as neutral rather than the correct class.

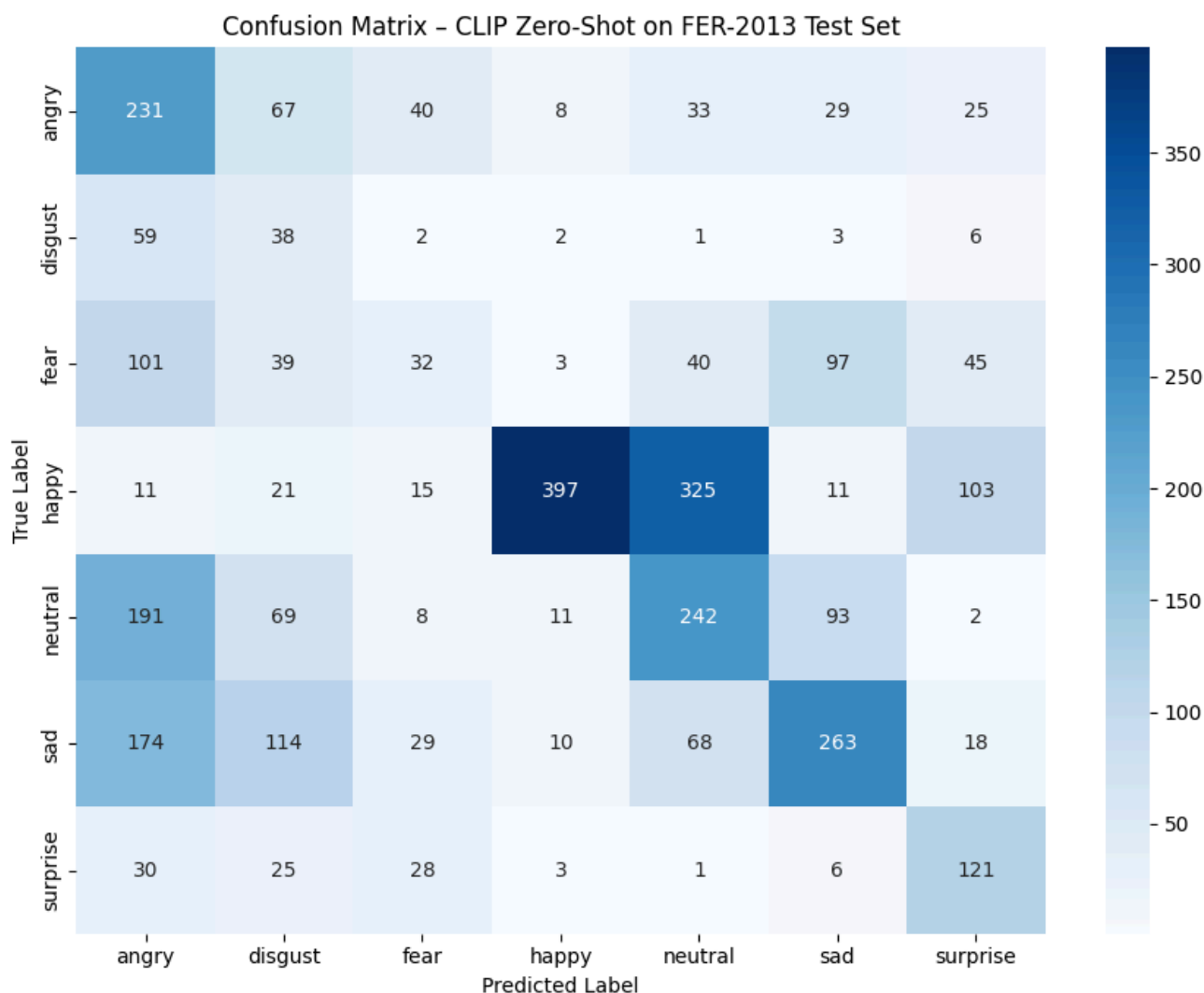
8.2 CLIP Confusion Matrix

To complement the qualitative similarity matrix analysis, the CLIP zero-shot classifier is evaluated quantitatively on the full FER-2013 test set of 3,290 images. For each test image, the same prompt-based classification pipeline is applied and the predicted class is compared against the ground truth label. This allows a direct, class-level comparison between CLIP zero-shot and the fine-tuned ResNet18.

CLIP evaluation: 100%|██████████| 52/52 [00:39<00:00, 1.31it/s]

CLIP Zero-Shot — Classification Report

	precision	recall	f1-score	support
angry	0.29	0.53	0.38	433
disgust	0.10	0.34	0.16	111
fear	0.21	0.09	0.13	357
happy	0.91	0.45	0.60	883
neutral	0.34	0.39	0.37	616
sad	0.52	0.39	0.45	676
surprise	0.38	0.57	0.45	214
accuracy			0.40	3290
macro avg	0.39	0.39	0.36	3290
weighted avg	0.51	0.40	0.42	3290



The model achieves an overall accuracy of 40% on the full test set of 3,290 images. The most striking result concerns happiness. Despite a precision of 0.91, the recall is only 0.45. This means that, while CLIP is very confident in its predictions of 'happy', it incorrectly classifies more than half of all 'happy' images. 325 of them are mislabelled as 'neutral'. This classic precision-recall trade-off is caused by CLIP's tendency to predict 'neutral' as a safe default for ambiguous expressions.

Fear is the weakest class with an F1-score of only 0.13 and a recall of 9%. The confusion matrix confirms that fear images are scattered across angry (101), sad (97), and neutral (40).

Disgust similarly struggles with an F1-score of 0.16, with 59 images misclassified as angry.

Angry and surprise perform moderately with recalls of 53% and 57% respectively, benefiting from their visually distinctive features. Sad and neutral show moderate F1-scores around 0.37–0.45 but are heavily confused with each other and with angry.

9. Generalisation Test — Own Photos

To evaluate out-of-distribution generalisation, the model is tested on real photographs of the project team members. This constitutes a **domain shift**: FER-2013 consists of low-resolution (48×48), grayscale, web-scraped images, while the team photos are high-resolution colour images taken under natural lighting.

Preprocessing pipeline for own photos:

1. Face detection using OpenCV Haar Cascade (`haarcascade_frontalface_default.xml`)
2. Face crop with 10% padding to include forehead and chin context
3. Conversion to grayscale, then to 3-channel RGB (matching training format)
4. Resize to 224×224, ImageNet normalisation

Upload your photo (JPG/PNG):

Keine ausgewählt

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving Foto.jpg to Foto (2).jpg

Face detected!

Foto (2).jpg:

angry	0.6%
disgust	0.0%
fear	0.1%
happy	96.1%
neutral	0.6%
sad	0.0%
surprise	2.7%

Predicted: happy
(96.1% confidence)



10. Conclusion

In this project we tackled facial expression recognition on the FER-2013 dataset — classifying faces into 7 emotion classes (angry, disgust, fear, happy, neutral, sad, surprise) — and compared a fine-tuned ResNet18 CNN against a CLIP zero-shot baseline.

Setup:

- ResNet18 pre-trained on ImageNet; backbone frozen, only layer4 + a new classification head (Dropout 0.5) fine-tuned.
- Trained in two phases (10 + 10 epochs), Adam (lr 1e-4, weight decay 1e-4), ReduceLROnPlateau scheduler, batch size 64, with data augmentation.

Key results:

- Best validation accuracy: ~61% (59.2% after phase 1, 61.4% after the 10-epoch extension)
- Test accuracy: ~62% (61.8%) on the held-out test set
- CLIP zero-shot baseline: ~40% (40.5%) — without any task-specific training
- Fine-tuning advantage over CLIP zero-shot: +21 percentage points
- For reference, human performance on FER-2013 has been reported at around 65%, while stronger recent models reach roughly the mid-70% range.

Observations:

- The second training phase confirmed the plateau: validation accuracy rose only marginally from ~59% to ~61%, while training accuracy climbed to ~89% — a clear overfitting signal that Dropout(0.5) and weight decay reduced but did not eliminate. The best checkpoint was saved at the validation peak.
- Per-class performance reflects the class imbalance: 'happy' is recognised best (F1 0.85, the class with by far the most training data), while 'fear' is weakest (F1 0.37) and the minority class 'disgust' remains comparatively low at F1 0.59, with only 436 images in the original FER-2013 training split. Errors concentrate on the visually similar negative emotions (fear / sad / angry) rather than being random.
- Feature maps confirm the transfer-learning logic: layer1 detects generic edges and textures that look the same across all emotions (so it can stay frozen), while layer4 produces sparse, emotion-specific activations (so it benefits most from fine-tuning).
- CLIP reaching ~40% with zero labelled examples shows how strong general-purpose vision-language models are as baselines — though it clearly trails the supervised model and, in the one-sample-per-class qualitative check, classified 4 of 7 examples correctly. The main confusions occurred for disgust, fear and sad.
- Out-of-distribution testing on our own photos revealed a clear domain gap: the model, trained on 48×48 grayscale web-scraped images, struggles with high-resolution colour photos. Closing this gap would require face alignment, a more diverse / balanced dataset, and likely a larger backbone such as ResNet50 or EfficientNet.

11. Limitations & Potential Improvements

- **Weighted cross-entropy loss** or **oversampling** for underrepresented classes (disgust, fear) would likely improve per-class recall significantly.
- **Face alignment** (e.g. MTCNN) prior to training would normalise pose variation and likely improve generalisation.
- **Larger backbone** (ResNet50, EfficientNet) with more trainable parameters could capture more nuanced facial features.

- **Full fine-tuning** with a lower learning rate and longer training schedule may close the remaining gap to SOTA (~75%).
- **Training on colour images** or augmenting with colour channels would reduce the domain gap for real-world deployment.

12. Discloser of GenAI

Generative AI tools (specifically Claude) were used in a supportive manner throughout the development of this project. GenAI was primarily used to assist with debugging and code refinement, fixing syntax and runtime errors, and improving the structure and wording of report text and the speaker script for the presentation. All substantial aspects of the project were carried out independently, including the model architecture decisions, the transfer learning strategy, the choice of hyperparameters, the interpretation of training curves, confusion matrices and feature map visualisations, as well as the overall analysis and conclusions drawn from the results.